Reg. No. | | | | | | | | | | | [13AE662/12AE662]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Solve the linear programming problem by graphical method: **04**

Max Z $=3x_1+2x_2$ subject to constraints

$$x_1 - x_2 \leq 1$$

$$x_1 + x_2 \geq 3,\ x_1, x_2 \geq 0$$

b. The ABC company has been a producer of picture tubes for television sets and certain printed circuits for radios. The company has just expanded into full scale production and marketing of AM and AM-FM radios. It has built a new plant that con operate 48 hours per week. Production of an AM radio in the new plant will require 2 hours and production of an AM-FM radio will require 3 hours. Each AM radio will contribute Rs 40 to profits while an AM-FM radio will contribute Rs.80 to profits. The marketing department, after extensive research has determined that a maximum of 15 AM radios and 10 AM-FM radios can be sold each week. **07**

i. Formulate a LP model to determine the optimum production mix of AM and AM-FM radios that will maximize profits.

ii. Solve the above problem using graphical method

c. Mention the different phases of an operations research study **03**

2. a. Define feasible solution of linear programming problem. Use simplex method to Maximize Z $=2x_1+3x_2$. Subject to the constraints, $2x_1+x_2 \leq 12$ **07**

$$x_1+3x_2 \leq 15,\ x_1, x_2 \geq 0$$

b. Apply Big-M method, to solve LPP. Minimize Z $=2x_1+x_2$ subject to **07**

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4, \quad x_1, x_2 >= 0$$

3. a. Write the dual of following LPP: Minimize $Z = 2x_2 + 8x_3$ subject to **04**

$$3x_1 + x_2 \geq 12$$
$$2x_1 + x_2 + 6x_3 \leq 6$$
$$5x_1 - x_2 + 3x_3 = 4$$
$$x_1, x_2, x_3 \geq 0$$

b. Solve the LPP by dual simplex method, minimize $Z = 2x_1 + x_2$, subject to the constraints $3x_1 + x_2 \geq 3,\ 4x_1 + 3x_2 \geq 6, x_1 + 2x_2 \geq 3\ \ and\ x_1; x_2 \geq 0$ **10**

4. a. Obtain optimum feasible solution to the transportation problem by finding initial basic feasible solution using least count entry method:

DESTINATIONS						Available (supply)
		D_1	D_2	D_3	D_4	
ORIGIN	O_1	1	2	1	4	30
	O_2	3	3	2	1	50
	O_3	4	2	5	9	20
Required (demand)		20	40	30	10	

b. A batch of four jobs can be assigned to five different machines. The following table gives the time for each job on various machines. Obtain the optimal assignment of jobs to machines, which will minimize the total time machines. **07**

		M_1	M_2	M_3	M_4	M_5
JOBS	J_1	10	11	4	2	8
	J_2	7	11	10	14	12
	J_3	5	6	9	12	14
	J_4	13	15	11	10	7

5. a. Use the revised simplex method to solve the following LP problem **07**

Maximize $Z = 2x_1 + x_2$

Subject to the constraints

$$3x_1 + 4x_2 \leq 6$$
$$6x_1 + x_2 \leq 3$$

b. Solve integer programming problem, **07**

Max $Z = 7x_1 + 9x_2$, subject to $-x_1 + 3x_2 \leq 7$

$7x_1 + x_2 \leq 35,\ x_1 \geq 0, x_2 \geq 0$ and integers

6. a. Two players A and B simultaneously show 2 or 3 fingers. If the sum of fingers shows even then A wins the sum from B. if the sum is odd then A looses the sum to B. Determine the optimal strategies for each player and value of game by formulating the pay-off matrix for player A. **04**

b. Solve the following game by using the principle of dominance given the pay-off matrix for player A. **05**

B

		B_1	B_2	B_3	B_4	B_5
A	A_1	4	4	2	-4	-6
	A_2	8	6	8	-4	0
	A_3	10	2	4	10	12

determine optimal strategies for each player and value of game for player A

c. Solve the following (2x3) game graphically: **05**

B

		B_1	B_2	B_3
A	A_1	3	-1	0
	A_2	2	1	-1

7. a. The following table gives the list of activities involved in a project and duration: **07**

Activity	1-2	1-3	1-4	2-5	3-4	3-7	4-5	4-6	5-6	4-7	6-7
Duration (days)	20	24	08	20	16	24	0	18	0	4	12

i. Draw the network diagram for project and determine its critical path.

ii. Obtain the float for each activity in project.

b. A project is composed of seven activities whose time estimates are given below: **07**

Activity	Estimated duration (weeks)		
	Optimistic	Pessimistic	Most likely
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

i. Draw the project network, expected duration and variance of each activity.

ii. Determine the critical path and expected project length

iii. What is the probability that project will be completed within 21 weeks.

8. a. Explain characteristics of queueing system. 04

b. The arrival of customers at a banking counter follows poisson distribution with a mean of 45 per hour. The service rate of the counter clerk also follows poisson distribution with a mean of 60 per hour. 05

i. What is the probability of having zero customers in the system (P_o)?

ii. Determine the steady state performance statistics namely L_s L_q, W_s and W_q

c. An airline company has one reservation clerk on duty in their local branch at any given time. The clerk handles information regarding passenger reservation and flight timings. Assume that number of customers arriving during any given period is poisson distributed with an arrival rate of eight per hour and that reservation clerk can serve a customer in six minutes on an average, with an exponentially distributed service time. 05

i. What is the probability that the system is busy?

ii. What is the average time a customers spends in the system?

iii. What is the average length of queue and what is number of customers in system?